

TTC Domains

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Object Reallocation Problem

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- agent i owns object o_i
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Goal: Identify rules that satisfy desirable properties.

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- ① individually rational, Pareto efficient, and strategyproof (Ma [1994])
- ② individually rational, pair efficient, and strategyproof (Ekici [2024])

A problem with TTC

- TTC outcomes may not always be desirable:

P_1	P_2	P_3	$\dots$	P_k	$\dots$	P_{n-1}	P_n
o_2	o_1	o_2	$\dots$	o_{k-1}	$\dots$	o_{n-2}	o_{n-1}
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Red: TTC allocation

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Green: an alternative desirable allocation

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o_n	o_2	o_3	$\dots$	o_k	$\dots$	o_{n-1}	o_n
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Question: Do the characterizations of TTC extend to restricted domains?

This paper

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- Result 1: If a domain satisfies the top-two condition, it is a TTC domain.
- Result 2: If a domain fails the top-two condition [in a certain way], it is not a TTC domain.

Model

- $N = \{1, \dots, n\}$: set of agents
- $O = \{o_1, \dots, o_n\}$: set of objects, o_i is agent i 's endowment
- $\mathcal{D} \subset \mathcal{P}$: the preference domain (subset of strict linear orders over O)
- $\mathcal{X}$: set of allocations ($x : N \rightarrow O$)
- A rule $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$ associates with each preference profile $P \in \mathcal{D}^N$ an allocation $x \in \mathcal{X}$.

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Notation:

- $r_k(P_0, O')$: object ranked k -th by P_0 within O'
- $r_k(\mathcal{D}', O')$: objects that are ranked k -th by some $P_0 \in \mathcal{D}'$ within O'

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- ② $\mathcal{D}_2 = \{o_1 o_2 o_3, o_2 o_3 o_1, o_1 o_3 o_2\}$ ✗
- ③ $\mathcal{D}_3 = \{o_1 o_2 o_3 o_4, o_1 o_3 o_2 o_4, o_2 o_1 o_4 o_3, o_2 o_4 o_3 o_1\}$

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- ③ $\mathcal{D}_3 = \{o_1 o_2 o_3 o_4, o_1 o_3 o_2 o_4, o_2 o_1 o_4 o_3, o_2 o_4 o_3 o_1\}$ ✗

Top-two is sufficient

Theorem 1.

Suppose $\mathcal{D} \subset \mathcal{P}$ satisfies the top-two condition. Let $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$ be a rule. The following are equivalent:

- (i) φ is the TTC rule.*
- (ii) φ is individually rational, Pareto efficient, and strategyproof.*
- (iii) φ is individually rational, pair efficient, and strategyproof.*

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It follows that the following are TTC domains:

- single-dipped domain (Tamura [2023], Hu and Zhang [2024])
- every partial-agreement domain

$$\mathcal{D}^{PA}(\succ) = \{P_0 \in \mathcal{P} : \forall a, b \in O, a \succ b \Rightarrow a P_0 b\}$$

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- Consider $(P'_S, P_{-S}) \in \mathcal{D}^N$ where:

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- Since φ is IR, either
 - 1 $\varphi_i(P'_S, P_{-S}) = o_{i+1}$ for all $i \in S$
 - 2 $\varphi_i(P'_S, P_{-S}) = o_i$ for all $i \in S$.

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- IR and SP $\implies \varphi$ must execute cycle S at P as well.

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Existing proofs on $\mathcal{D} = \mathcal{P}$: Ekici (2024), Ekici and Sethuraman (2024)

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- The proof of Theorem 1 shows that any such rule must execute TTC cycles whenever the required auxiliary preferences can be constructed.
- Hence, a deviation from TTC is possible only when those auxiliary preferences cannot be constructed.
- We construct such a rule for a broad subclass of domains that fail the top-two condition.

Necessity

Theorem 2.

Suppose $\mathcal{D} \subset \mathcal{P}$ does not satisfy the top-two condition. Suppose further that there exists a subset $O' \subset O$ such that:

- (i) $\mathcal{D}$ does not satisfy the top-two condition within O' ;*
- (ii) $|O'| \leq 4$;*
- (iii) for every $o \in O \setminus O'$, $o \in r_1(\mathcal{D}, O' \cup \{o\})$.*

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- circular domain

Proof

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- A non-TTC rule on this domain proves necessity of the top-two condition.

- We introduce the top-two condition and establish it as a valuable criterion for studying object reallocation under restricted domains.
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Thank you!